

R4 Newton's third law

Name: _____

Class: _____

Date: _____

Time: **395 minutes**

Marks: **323 marks**

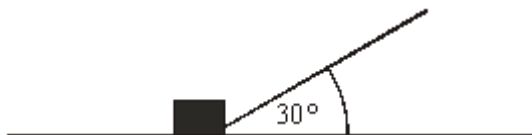
Comments:

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Q1.

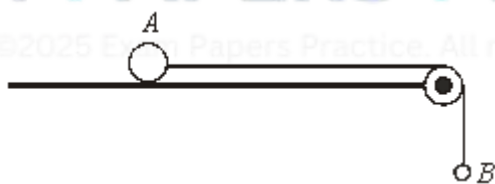
The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2} .



- (a) Draw a diagram to show all of the forces acting on the box. (1)
 - (b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures. (3)
 - (c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N. (3)
 - (d) Find the coefficient of friction between the box and the ground. (2)
- (Total 9 marks)**

Q2.

The diagram shows two particles, *A* and *B*, that are connected by a taut light inextensible string. Particle *A* has mass 8 kg and is on a rough horizontal surface. Particle *B* has mass 7 kg and is attached to the other end of the string, as shown in the diagram. The string passes over a smooth light pulley. Particle *A* moves towards the pulley as particle *B* falls.

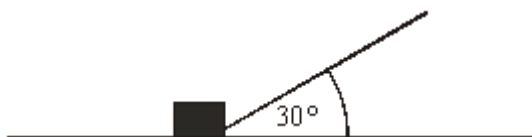


The coefficient of friction between *A* and the horizontal surface is 0.8.

- (a) Find the magnitude of the friction force between *A* and the surface. (2)
 - (b) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 0.392 m s^{-2} . (5)
- (Total 7 marks)**

Q3.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2} .

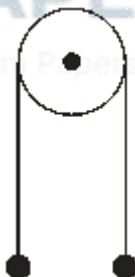


- (a) Draw a diagram to show all of the forces acting on the box. (1)
- (b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures. (3)
- (c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N. (3)
- (d) Find the coefficient of friction between the box and the ground. (2)
- (e) State what would happen to the magnitude of the friction force if the angle between the rope and the horizontal were increased. Give a reason for your answer. (2)

(Total 11 marks)

Q4.

Two particles are connected by a string, which passes over a pulley. Model the string as light and inextensible. The particles have masses of 2 kg and 5 kg. The particles are released from rest.

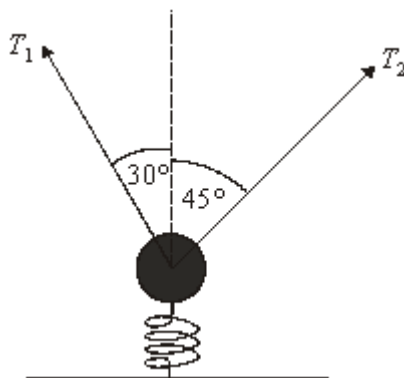


- (a) State one modelling assumption that you should make about the pulley in order to determine the acceleration of the particles. (1)
- (b) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 4.2 m s^{-2} . (5)
- (c) Find the tension in the string. (2)

(Total 8 marks)

Q5.

Two ropes are attached to a load of mass 500 kg. The ropes make angles of 30° and 45° to the vertical, as shown in the diagram. The tensions in these ropes are T_1 and T_2 newtons. The load is also supported by a vertical spring.



The system is in equilibrium and $T_1 = 200$.

- (a) Show that $T_2 = 141$, correct to three significant figures.

(3)

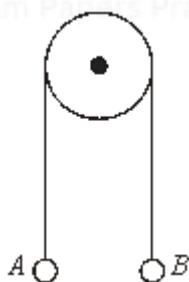
- (b) Find the force that the spring exerts on the load.

(4)

(Total 7 marks)

Q6.

Two particles, A of mass 5 kg and B of mass 9 kg, are attached to the ends of a light inextensible string. The string passes over a light smooth pulley as shown in the diagram. The particles are released from rest at the same height.



- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.8 m s^{-2} .

(5)

- (b) Find the tension in the string.

(2)

- (c) When B has been moving for 0.5 seconds it hits the floor. Find the height of A ,

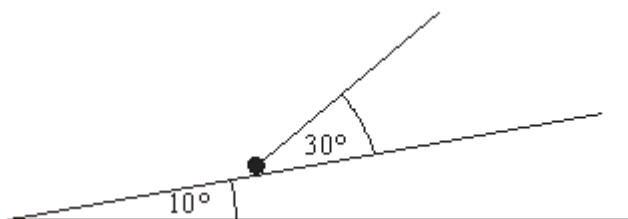
above the floor, at this time. Assume that A is still below the pulley when B hits the floor.

(4)

(Total 11 marks)

Q7.

A rough slope is inclined at an angle of 10° to the horizontal. A particle of mass 6 kg is on the slope. A string is attached to the particle and is at an angle of 30° to the slope. The tension in the string is 20 N . The diagram shows the slope, the particle and the string.



The particle moves up the slope with an acceleration of 0.4 m s^{-2} .

- (a) Draw a diagram to show the forces acting on the particle.

(1)

- (b) Show that the magnitude of the normal reaction force is 47.9 N , correct to three significant figures.

(4)

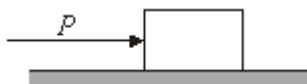
- (c) Find the coefficient of friction between the particle and the slope.

(6)

(Total 11 marks)

Q8.

A small block, of mass 2 kg , is on a rough horizontal surface. The coefficient of friction between the block and the surface is $\frac{1}{7}$. A horizontal force of magnitude P newtons is applied to the block, as shown in the diagram.



- (a) The magnitude of the frictional force that acts between the block and the surface is F newtons. Show that the maximum possible value of F is 2.8 .

(2)

- (b) The value of P increases gradually from 0 to 5 .

- (i) When $P = 1$, state the value of F .

(1)

- (ii) When $P = 5$, the block moves with constant acceleration $a\text{ m s}^{-2}$. Find the

value of a .

(4)

(Total 7 marks)

Q9.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

- (a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.

(i) Draw a diagram showing the forces acting on the car.

(2)

(ii) Show that $R = 660$.

(4)

(iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

(1)

- (b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

(i) Find the magnitude of the retardation of the car in this stage.

(2)

(ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

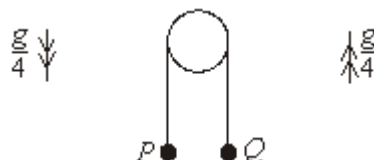
(2)

(Total 11 marks)

Q10.

Two particles, P and Q , are attached to the ends of a light inextensible string which hangs over a smooth fixed peg, as shown in the diagram. The masses of P and Q are $5m$ and km respectively, where $k < 5$.

The particles are released from rest and move with acceleration of magnitude $\frac{g}{4}$.



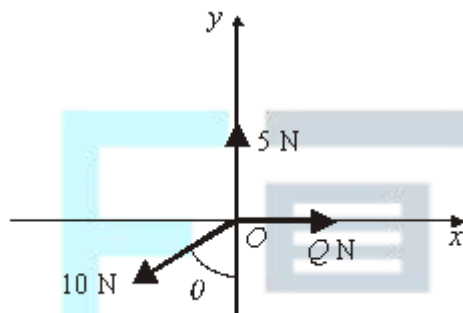
- (a) Find, in terms of m and g , the tension in the string.

(3)

- (b) Find the value of k . (3)
- (c) Find the magnitude of the force on the peg due to the string. (1)
- (d) The particles are released from rest at the same horizontal level when $t = 0$.
 Find, in terms of g , the vertical distance between the particles when $t = \frac{2}{3}$. (3)
- (Total 10 marks)**

Q11.

A particle rests in equilibrium at a point O on a horizontal surface. It is acted on by three horizontal forces of magnitudes 5 N, 10 N and Q N. The directions of these forces, relative to the horizontal axes Ox and Oy , are shown in the diagram below.



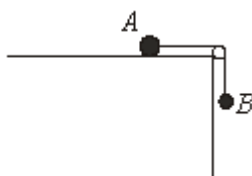
- (a) Show that $\theta = 60^\circ$. (2)
- (b) Find the value of Q . (2)
- (Total 4 marks)**

Q12.

A particle A of mass 0.2 kg is connected to a particle B of mass 0.3 kg by a light inextensible string.

The particle A is placed on a rough horizontal surface.

The string passes over a smooth fixed peg with B hanging freely, as shown in the diagram.



- (a) The system is held in equilibrium by a horizontal force, P newtons, which is applied to the particle A . The coefficient of friction between A and the rough horizontal surface is 0.5.
- (i) Draw a diagram showing the forces acting on the particle A . (2)
- (ii) Show that the magnitude of the maximum possible frictional force between A and the rough horizontal surface is 0.98 newtons. (2)
- (iii) Find the tension in the string. (2)
- (iv) Find the least value of P required to hold the system in equilibrium. (2)
- (b) The horizontal force P is removed and the particles start to move from rest.
- (i) Find the magnitude of the acceleration of the particles during the subsequent motion. (5)
- (ii) After falling 0.1 metres, the particle B hits the ground. Find the time between the system being released and B hitting the ground. (2)

(Total 15 marks)

Q13.

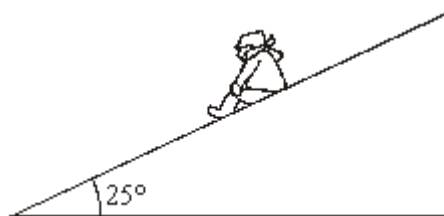
Four forces, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ newtons, $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ newtons, $\begin{bmatrix} -3 \\ -4.5 \end{bmatrix}$ newtons and $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ newtons, act on a particle.

- (a) Express the resultant, $\mathbf{F}$ newtons, of these four forces as a column vector. (2)
- (b) Find the magnitude of $\mathbf{F}$. (2)

(Total 4 marks)

Q14.

A children's slide is straight and inclined at 25° to the horizontal, as shown in the diagram.



Matthew, of mass 35 kg, goes down the slide at constant speed.

- (a) Draw a diagram to show the forces acting on Matthew. (1)
- (b) Find the magnitude of the normal reaction force between Matthew and the slide. (3)
- (c) Find the coefficient of friction between Matthew and the slide. (4)

(Total 8 marks)

Q15.

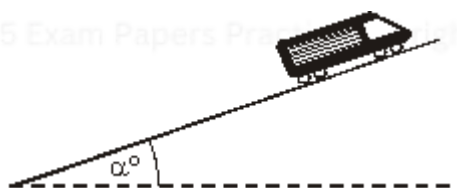
A mountain railway train moves on a straight track. The mass of the train and its passengers is 1000 kg. During its motion the train moves under the action of a variable propulsive force, P newtons, and a constant resistance force of R newtons.

- (a) During the first stage of its motion, the train moves horizontally with acceleration 0.25 m s^{-2} . In this stage, the value of P is 1200.

Show that $R = 950$.

(3)

- (b) During the second stage of its motion, the train moves up a slope inclined at an angle α to the horizontal, where $\sin \alpha = 0.1$.



In this stage, the value of $P = 2100$, and the value of R remains at 950. Find the acceleration of the train.

(4)

(Total 7 marks)

Q16.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle A , of mass 0.6 kg , is in contact with a smooth horizontal surface, and the particle B , of mass 0.1 kg , hangs freely above the ground. The system is released from rest with the string taut and A moves towards the peg.

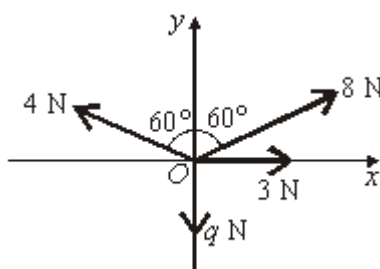
It can be assumed that, during the subsequent motion, A does **not** reach the peg.

- (a) While the particles move freely, the string is taut.
- (i) Show that the acceleration of the particles is 1.4 m s^{-2} . (5)
- (ii) Find the tension in the string. (1)
- (iii) Find the magnitude of the resultant force on the peg due to the tension in the string. (3)
- (b) After q seconds of the motion, the particle B hits the ground and remains there. The string connecting A and B slackens and A continues to move towards the peg. Sketch a velocity–time graph to show the two stages of the motion of A after being released from rest. (3)

(Total 12 marks)

Q17.

A particle is at a point O on a smooth horizontal surface. It is acted on by four horizontal forces of magnitudes 3 N , 8 N , 4 N and $q \text{ N}$. The directions of these forces, relative to the horizontal axes Ox and Oy , are shown in the diagram below.



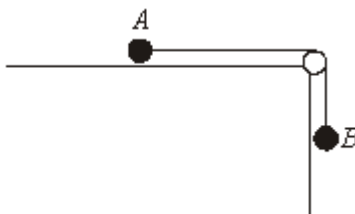
The resultant, $\mathbf{R}$ newtons, of these forces acts along the line Ox .

- (a) Show that $q = 6$. (4)
- (b) Find the magnitude of $\mathbf{R}$.

(3)
(Total 7 marks)

Q18.

Two particles, A and B , are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



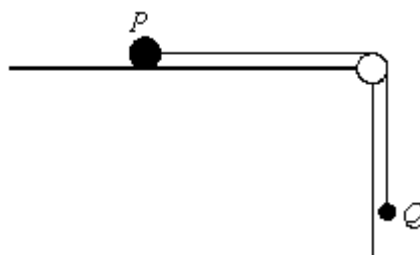
The particle A is of mass 0.5 kg and the particle B is of mass 0.1 kg . The particle A is in contact with a rough horizontal surface.

- (a) The system is at rest and the particle B hangs vertically.
- (i) Show that the tension in the string is 0.98 newtons . (1)
 - (ii) The particle A rests in limiting equilibrium. Show that the coefficient of friction between A and the surface is 0.2 . (3)
 - (iii) Draw a diagram to show the forces acting on the peg due to the string. (1)
 - (iv) Find the magnitude of the resultant force on the peg due to the string. (2)
- (b) An additional mass of 0.1 kg is attached to the particle B and the system is released from rest. The particle A moves across the surface towards the peg and B moves vertically downwards. The particles move with acceleration $a \text{ m s}^{-2}$ and the tension in the string is $T \text{ newtons}$.
- (i) Show that $a = 1.4$. (5)
 - (ii) Find the value of T . (1)
 - (iii) After falling a distance of 0.7 metres , B hits the ground. Find the time between the particles being released and B hitting the ground. (2)

(2)
(Total 15 marks)

Q19.

Two particles, P and Q , are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle P is of mass $2m$ and Q is of mass m . The particle P is in contact with a horizontal surface, which may be modelled as either rough or smooth.

- (a) In the first case, the surface is modelled as rough and the particle Q hangs at rest. The coefficient of friction between P and the surface is μ .
- (i) Find the tension in the string.
- (ii) Find the range of possible values of μ .
- (b) In the second case, the surface is modelled as smooth. The system is released from rest with the string taut and the particle P moves towards the peg.
- Find the tension in the string.

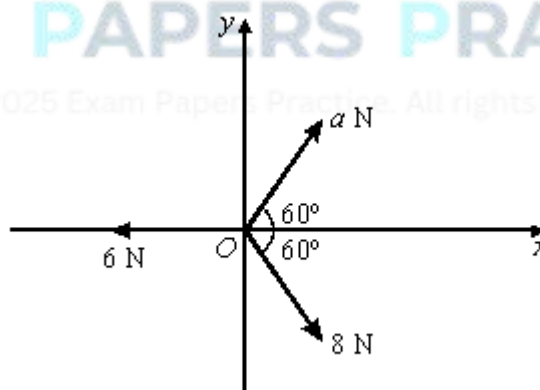
(1)

(3)

(5)

(Total 9 marks)

Q20.



A particle is at a point O on a smooth horizontal surface. It is acted on by three horizontal forces of magnitudes 6 N, 8 N and α N. Relative to horizontal axes Ox and Oy , the directions of these three forces are shown in the diagram. The resultant, $\mathbf{R}$, of these forces acts along the line Oy .

- (a) Show that $\alpha = 4$.

(4)

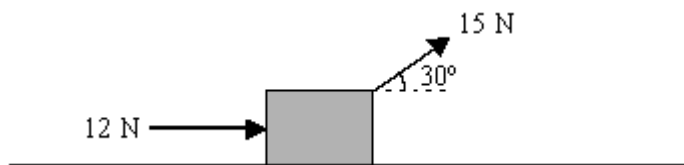
- (b) Find the magnitude of R .

(3)

(Total 7 marks)

Q21.

The diagram shows a small box resting on a rough horizontal surface.



The box is of weight W newtons. It is pushed with a horizontal force of 12 newtons, and pulled with a force of 15 newtons at an angle of 30° to the horizontal.

The box rests in limiting equilibrium.

- (a) Draw a diagram to show all the forces acting on the box.

(2)

- (b) Show that the frictional force acting on the box is approximately 25 newtons.

(3)

- (c) The coefficient of friction between the box and the surface is $\frac{1}{3}$.
Find the normal reaction force between the box and the surface.

(2)

- (d) Find the value of W .

(3)

(Total 10 marks)

Q22.

The diagram shows a car pulling a trailer in a straight line on a horizontal stretch of road.



The mass of the car is 1250 kg and the total resistance force acting on the car is 500 N.

The mass of the trailer is 250 kg and the total resistance force acting on the trailer is 100 N.

- (a) During part of the journey, a constant braking force is applied to the car, causing the car and trailer to decelerate at a constant rate of 0.5 m s^{-2} .
- (i) By considering the forces on the trailer, find the magnitude of the force in the towbar between the car and the trailer.

(3)

- (ii) Find the magnitude of the braking force applied to the car.

(4)

- (b) Later in the journey, the car and trailer travel with constant speed.

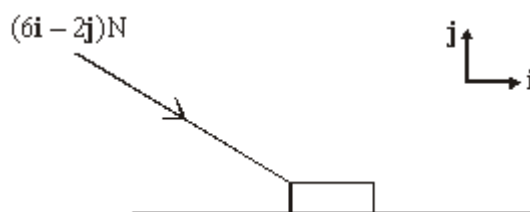
State the magnitude of the tension in the towbar.

(1)

(Total 8 marks)

Q23.

A small case is moving on a rough horizontal surface. The case is being pushed with a force $(6\mathbf{i} - 2\mathbf{j})$ newtons, as shown in the diagram.



The case has mass 1.5 kg and moves with acceleration 0.6 m s^{-2} .

- (a) Draw a diagram to show the forces acting on the case.

(1)

- (b) Show that the frictional force acting on the case is of magnitude 5.1 newtons.

(3)

- (c) Find the normal reaction force between the case and the surface.

(3)

- (d) The coefficient of friction between the case and the surface is μ . Find the value of μ .

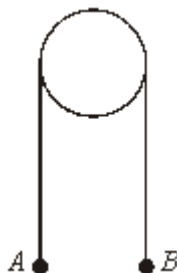
(2)

(Total 9 marks)

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Q24.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth fixed peg, as shown in the diagram.



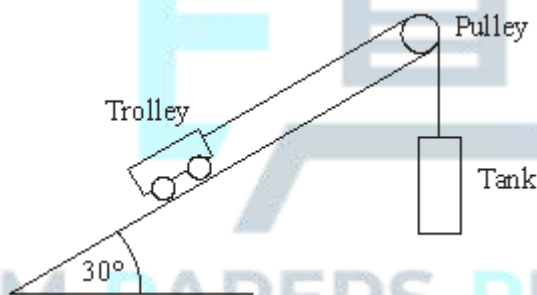
The particles A and B have masses 0.21 kg and 0.14 kg respectively. The system is released from rest. It may be assumed that, during the motion, neither particle reaches the

peg.

- (a) Show that the magnitude of the acceleration of the particles during the subsequent motion is 1.96 m s^{-2} . (5)
- (b) Find the speed of the particles after 2 seconds of the motion. (2)
- (c) After 2 seconds of the motion, the string breaks and the particles move independently under the influence of gravity alone. Find the time between the string breaking and the particle B coming to instantaneous rest. (2)
- (d) Sketch a velocity-time graph to show the motion of B between being released from rest and subsequently coming instantaneously to rest. (3)
- (Total 12 marks)**

Q25.

A trolley, of mass 200 kg , is on a slope inclined at 30° to the horizontal. It is attached to a tank by a light, inextensible rope that passes over a smooth, light pulley. The tank can be filled with water. The situation is shown in the diagram below.



Model the tank and trolley as particles and assume that there is no resistance to motion.

- (a) When the tank is filled with water, the total mass of the tank and the water is 150 kg .
Show that the acceleration of the tank and trolley is 1.4 m s^{-2} . (7)
- (b) When the tank is empty, the trolley can be set in motion, so that it moves with a constant speed up the slope. Find the mass of the empty tank. (4)
- (Total 11 marks)**

Q26.

A particle, of mass 2 kg , is set into motion up a rough slope inclined at 40° to the horizontal.

The coefficient of friction between the particle and the slope is 0.3 . Assume that there is no air resistance acting on the particle.

- (a) Find the magnitude of the friction force that acts on the particle. (3)
- (b) Find the acceleration of the particle as it moves up the slope. (3)
- (Total 6 marks)

Q27.

A lift has mass 800 kg. A vertical cable is attached to the lift. Model the lift as a particle.

- (a) Show that the tension in the cable is 8000 N, when the lift is accelerating upwards at 0.2 m s^{-2} . (3)
- (b) Find the tension in the cable when the lift is accelerating downwards at 0.2 m s^{-2} . (2)
- (c) Write down the tension in the cable when the lift is travelling at a constant speed. (1)
- (Total 6 marks)

Q28.

The diagram shows two particles, A and B , which are connected by a light inextensible string. The string passes over a smooth light pulley. Particle A has mass 20 kg and rests on a rough horizontal surface. The coefficient of friction between the particle and the surface is 0.15. Particle B has mass 5 kg and hangs with the string vertical, as shown in the diagram.



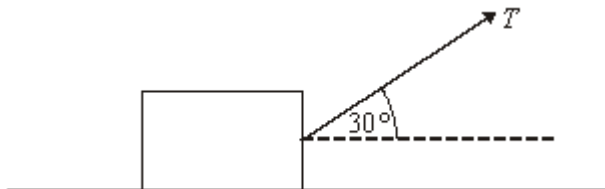
The particles are released from rest and begin to move.

- (a) Find the magnitude of the friction force acting on A . (2)
- (b) By forming an equation of motion for each particle, show that the acceleration of each particle has magnitude 0.784 m s^{-2} . (5)
- (c) Find the tension in the string. (2)
- (d) Find the distance that A moves in the first 3 seconds of its motion. Assume that the acceleration of A remains constant during this time.

(3)
(Total 12 marks)

Q29.

A heavy crate, of mass 200 kg, is pulled along a rough horizontal surface at a constant speed by a rope. The rope is at an angle of 30° to the horizontal. The tension in the rope is T newtons. The coefficient of friction between the crate and the surface is 0.6. Model the crate as a particle.

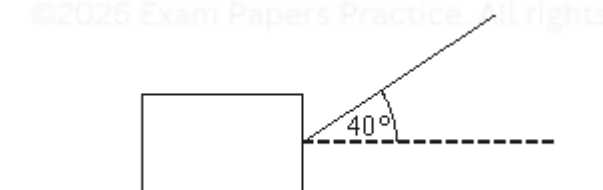


- (a) Draw a diagram to show the forces acting on the crate. (1)
- (b) Show that the magnitude of the normal reaction force on the crate is $(1960 - 0.5T)$ newtons. (3)
- (c) Find T . (5)

(Total 9 marks)

Q30.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is T newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.



Model the box as a particle.

- (a) Draw a diagram to show the forces acting on the box. (1)
- (b) Find, in terms of T , the magnitude of the normal reaction force acting on the box. (3)
- (c) Show that the magnitude of the friction force acting on the box is

$$294 - 0.6T \sin 40^\circ$$

(2)

- (d) The box accelerates at 0.5 m s^{-2} . Find T .

(4)

(Total 10 marks)

Q31.

A rough plane is inclined at an angle of 40° to the horizontal. A particle, of mass 5 kg , is sliding down the plane.

- (a) Draw a diagram to show the forces acting on the particle.

(1)

- (b) Find the magnitude of the normal reaction force acting on the particle.

(2)

- (c) The coefficient of friction between the particle and the plane is 0.2 . Show that the magnitude of the friction force acting on the particle is 7.51 N , correct to three significant figures.

(2)

- (d) Show that the acceleration of the particle is 4.80 m s^{-2} , correct to three significant figures.

(4)

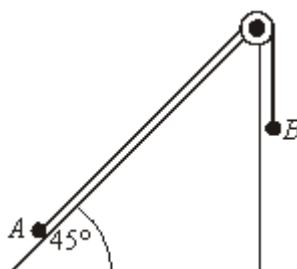
- (e) Find the distance that the particle travels as its speed increases from 2 m s^{-1} to 10 m s^{-1} .

(3)

(Total 12 marks)

Q32.

Two particles, A and B , are connected by a light inextensible string, which passes over a smooth light pulley. Particle A is on a smooth slope, at 45° to the horizontal, and particle B hangs with the string vertical, as shown in the diagram.



The mass of A is 14 kg and the mass of B is 6 kg .

- (a) Using two equations of motion, show that the acceleration of the particles is 1.91 m s^{-2} correct to three significant figures.

(6)

- (b) Particle B is replaced by a particle C of mass m kg. After the particles have been set in motion, they move with a constant speed. Find m .

(4)

(Total 10 marks)

Q33.

A crate, of mass 50 kg, is at rest on a warehouse floor. The floor is rough and horizontal. The coefficient of friction between the crate and the floor is μ . A rope is attached to the crate at an angle of 30° to the horizontal. The tension in the rope is 100 N. The crate is shown in the diagram.



Model the crate as a particle.

- (a) Draw and label a diagram to show the forces acting on the crate.

(1)

- (b) Show that the magnitude of the normal reaction force acting on the crate is 440 N.

(3)

- (c) If the crate remains at rest, μ must satisfy the inequality $\mu \geq k$. Find k .

(3)

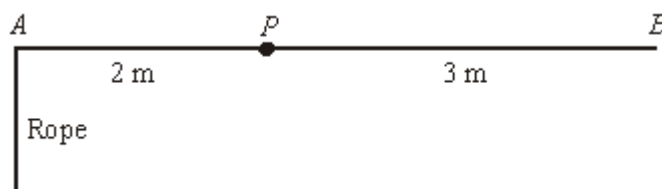
- (d) If $\mu = 0.1$, find the acceleration of the crate.

(4)

(Total 11 marks)

Q34.

A uniform beam AB has length 5 metres and mass 10 kg. It is freely pivoted at the point P which is 2 metres from A . A rope is attached to the beam at A and exerts a vertical force on the beam at A . The beam and rope are shown in the diagram.



- (a) The beam is in equilibrium and at rest in a horizontal position.

- (i) Show that the tension in the rope is 24.5 N.

(3)

- (ii) A particle of mass 40 kg is then fixed to the beam at B . The tension in the rope

is increased, so that the beam remains horizontal. Find the tension in the rope in this case.

(3)

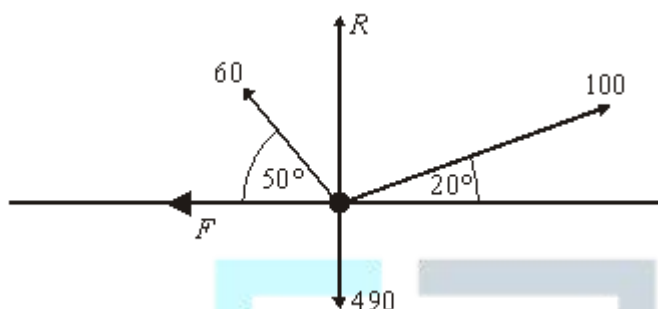
- (b) How would your answer to part (a)(ii) change if the beam had been in equilibrium at an angle to the horizontal, but with the rope still vertical? Explain why.

(2)

(Total 8 marks)

Q35.

A box, of mass 50 kg, moves in a straight line on a rough horizontal surface. The diagram below shows **all** the forces acting on the box, as it moves. The magnitude of each force is in newtons. The box is modelled as a particle.



Note that the weight of the box has been included in the diagram.

- (a) Find R , the magnitude of the normal reaction force acting on the box.

(3)

- (b) The box accelerates at 0.5 m s^{-2} .

- (i) Show that F , the magnitude of the friction force acting on the box, is approximately 30.4.

(4)

- (ii) Find the coefficient of friction between the box and the surface.

(2)

(Total 9 marks)